



Technical Data Sheet

M-610 Platinum cure mold making Silicone Rubber

Description

Platinum cure silicone also called Poly-addition silicones, are ideal silicone casting rubbers. They are excellent for moulding with a fully enclosed mould, as they cure at room temperature conditions. Platinum silicones are very stable. They have very low, or no shrinkage, we supply them in a wide range of hardnesses (Shore A) and they have very high tear strength, excellent mould and library life. M-610 silicone cure to soft, strong rubbers which are tear resistant and exhibit very low long term shrinkage. Moulds made with M-610 will last a long time in your mould library.

Features & Benefits

- *extremely low viscosity and high flowability
- *A kind of soft silicone to copy the finest patterns
- *Very accurate reproduction of dimensions (very low shrinkage)
- *Molds do not require release agents when casting most materials
- *Can be made thixotropic (nonflowable) for vertical surface replication
- *High-strength
- *Excellent chemical resistance
- *Ideal for the most demanding resin casting applications

Applications

M-610 is good for casting wax, gypsum, resins, concrete and other materials.

Technical Parameters

Typical general characteristics	Inspection Method	Value
Product data (Catalyzed A+B)		M-610
Hardness	Shore A	10±2
Mix Ratio	A:B	1:1 (or 10:1)
Color	/	translucent
Pot life	25 °C (77F)	30-40mins
Demold Time	25 °C (77F)	3-4 hours
Mixed viscosity	mPa.s	1500±300
Tear-strength	kg/cm	≥17
Tensile-strength	M.pa	≥3.5
Elongation	%	≥650
Shrinkage	%	≤0.1%

How To Use

Note: *This product will not cure against surfaces containing sulfur, even when sealed.*

Measuring & Mixing

Before you begin, pre-mix Part A and Part B separately. After dispensing required amounts of Parts A and B into mixing container (1A:1B by weight), mix thoroughly making sure that you scrape the sides and bottom of the mixing container several times. The rubber should be a uniform color with no streaks.

Optional... Vacuum Degassing

Although not necessary, vacuum degassing helps eliminate any entrapped air in pourable silicone rubber. After mixing parts A and B, vacuum material for 2-3 minutes at 29 inches of mercury, making sure that you leave enough room in container for product expansion.

Pouring

For best results, pour your mixture in a single spot at the lowest point of the containment field. Let the rubber seek its own level. A uniform flow will help minimize entrapped air. If using as a mold material, the liquid rubber should level off at least 1/2" (1.3 cm) over the highest point of the model surface.

Curing

Allow M-610 silicone rubber to cure for 4 hours at room temperature (73 ° F/23 ° C) before demolding.

Heat Curing - Time to demold can be reduced by applying mild heat. Example: After pouring M-610 rubber at room temperature, place the mold in a hot box or industrial oven at 140 ° F (60 ° C). This will reduce the time to demold of a 1/2" (1.3 cm) thick section to about 10 minutes.

Note: Time will vary depending on mold thickness.

Cure Inhibition

Addition cured silicone rubber may be inhibited by certain contaminants in or on the pattern to be molded resulting in tackiness at the pattern interface or a total lack of cure throughout the mold. Latex, sulfur clays, certain wood surfaces, newly cast polyester, epoxy or urethane rubber may cause inhibition. If compatibility between the rubber and the surface is a concern, a small-scale test is recommended. Apply a small amount of rubber onto a non-critical area of the pattern. Inhibition has occurred if the rubber is gummy or uncured after the recommended cure time has passed. To prevent inhibition, one or more coatings of a clear acrylic lacquer applied to the model surface is usually effective. Allow any sealer to thoroughly dry before applying rubber.

Appearance

Part A/B: Translucent liquid

ATT: We can adjust any color according to customer's requirements

Usable Life and Storage

Product should be stored at or below 32°C (89.6°F) in original, unopened containers, M-610 have a usable life of 12 months.

Package

1kg/drum; 20kg/drum;25kg/drum;200kg/drum.

